Dashboard


Hadoop
Overview
Datanodes
Datanode Volume Failures
Snapshot
Startup Progress
Utilities
Overview 'namenode:8020' (active)
Started: Fri Oct 16 19:09:37 +0530 2026
Version: 3.3.6, r1be78238728da9266a4f88195058f08fd012bf9c
Compiled: Sun Jun 18 13:52:00 +0530 2023 by ubuntu from (HEAD detached at release-3.3.6-RC1)
Cluster ID: CID-7bea1e65-95c3-4c56-99b7-eb8a7a092d1f
Block Pool ID: BP-976675928-172.19.0.2-1792157974915
Summary
Security is off.
Safemode is off.
1 files and directories, 0 blocks (0 replicated blocks, 0 erasure coded block groups) = 1 total filesystem object(s).
Heap Memory used 135.83 MB of 311 MB Heap Memory. Max Heap Memory is 1.69 GB.
Non Heap Memory used 48.96 MB of 50.38 MB Commited Non Heap Memory. Max Non Heap Memory is <unbounded>.
Configured Capacity: 1006.85 GB
Configured Remote Capacity: 0 B
DFS Used: 24 KB (0%)
Non DFS Used: 3.97 GB
DFS Remaining: 951.67 GB (94.52%)
Block Pool Used: 24 KB (0%)
DataNodes usages% (Min/Median/Max/stdDev): 0.00% / 0.00% / 0.00% / 0.00%
Live Nodes 1 (Decommissioned: 0, In Maintenance: 0)
Dead Nodes 0 (Decommissioned: 0, In Maintenance: 0)
Decommissioning Nodes 0
Entering Maintenance Nodes 0
Total Datanode Volume Failures 0 (0 B)
Number of Under-Replicated Blocks 0
Number of Blocks Pending Deletion (including replicas) 0
Block Deletion Start Time Fri Oct 16 19:09:37 +0530 2026
Last Checkpoint Time Fri Oct 16 19:09:35 +0530 2026
Enabled Erasure Coding Policies RS-6-3-1024k
NameNode Journal Status
Current transaction ID: 1
Journal Manager
State
FileJournalManager(root=/tmp/hadoop-hadoop/dfs/name)
EditLogFileOutputStream(/tmp/hadoop-hadoop/dfs/name/current/edits_inprogress_0000000000000000001)
NameNode Storage
Storage Directory
Type
State
/tmp/hadoop-hadoop/dfs/name
IMAGE_AND_EDITS
Active
DFS Storage Types
Storage Type
Configured Capacity
Capacity Used
Capacity Remaining
Block Pool Used
Nodes In Service
DISK
1006.85 GB
24 KB (0%)
951.67 GB (94.52%)
24 KB
1
Hadoop, 2023.

Demo: Local Count Demonstration

```
PS D:\Coding\Learning\CloudComputing\Assignment08> cat input.txt | python mapper.py
MapReduce,1
is,1
the,1
core,1
processing,1
engine,1
in,1
Apache,1
Hadoop,1
that,1
analyzes,1
massive,1
datasets,1
in,1
parallel,1
across,1
a,1
distributed,1
cluster.,1
PS D:\Coding\Learning\CloudComputing\Assignment08> cat input.txt | python mapper.py | Sort-Object | python reducer.py | select -first 20
a,1
across,1
analyzes,1
Apache,1
cluster.,1
core,1
datasets,1
distributed,1
engine,1
Hadoop,1
in,2
is,1
MapReduce,1
massive,1
parallel,1
processing,1
that,1
the,1
PS D:\Coding\Learning\CloudComputing\Assignment08>
```

Demo: Hadoop Count Demo

```
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker compose up -d
[+] up 5/5
 ✓ Network hadoop_mapreduce_default Created
 ✓ Container resourcemanager        Started
 ✓ Container nodemanager            Started
 ✓ Container datanode               Started
 ✓ Container namenode               Started
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker exec -it namenode bash
bash-4.2$ mkdir input
bash-4.2$ echo "Hello Docker" >input/file1.txt
bash-4.2$ echo "Hello Hadoop" >input/file2.txt
bash-4.2$ hdfs dfs -mkdir -p /user/root/input
bash-4.2$ hdfs dfs -put ./input/* /user/root/input
bash-4.2$ ls input
file1.txt  file2.txt
bash-4.2$ hdfs dfs -ls /user/root/input
Found 2 items
-rw-r--r--   1 hadoop supergroup         13 2026-10-16 13:48 /user/root/input/file1.txt
-rw-r--r--   1 hadoop supergroup         13 2026-10-16 13:48 /user/root/input/file2.txt
bash-4.2$
```

```
bash-4.2$ echo -e "input file1.txt:" ; hdfs dfs -cat /user/root/input/file1.txt
input file1.txt:
Hello Docker
bash-4.2$ echo -e "input file2.txt:" ; hdfs dfs -cat /user/root/input/file2.txt
input file2.txt:
Hello Hadoop
bash-4.2$ echo -e "wordcount output:" ; hdfs dfs -cat output/part-r-00000
wordcount output:
Docker  1
Hadoop  1
Hello   2
bash-4.2$
```

```
bash-4.2$ hadoop jar share/hadoop/mapreduce/hadoop-mapreduce-examples-3.3.6.jar wordcount /user/root/input output
2026-10-16 13:54:50 INFO  DefaultNoHARMFailoverProxyProvider:64 - Connecting to ResourceManager at resourcemanager/172
19.0.4:8032
2026-10-16 13:54:51 INFO  JobResourceUploader:907 - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop
.staging/job_1792157978563_0002
2026-10-16 13:54:51 INFO  FileInputFormat:300 - Total input files to process : 2
2026-10-16 13:54:51 INFO  JobSubmitter:202 - number of splits:2
2026-10-16 13:54:52 INFO  JobSubmitter:298 - Submitting tokens for job: job_1792157978563_0002
2026-10-16 13:54:52 INFO  JobSubmitter:299 - Executing with tokens: []
2026-10-16 13:54:52 INFO  Configuration:2854 - resource-types.xml not found
2026-10-16 13:54:52 INFO  ResourceUtils:476 - Unable to find 'resource-types.xml'.
2026-10-16 13:54:52 INFO  YarnClientImpl:338 - Submitted application application_1792157978563_0002
2026-10-16 13:54:52 INFO  Job:1682 - The url to track the job: http://resourcemanager:8088/proxy/application_179215797
563_0002/
2026-10-16 13:54:52 INFO  Job:1727 - Running job: job_1792157978563_0002
2026-10-16 13:55:01 INFO  Job:1748 - Job job_1792157978563_0002 running in uber mode : false
2026-10-16 13:55:01 INFO  Job:1755 -  map 0% reduce 0%
2026-10-16 13:55:07 INFO  Job:1755 -  map 50% reduce 0%
2026-10-16 13:55:08 INFO  Job:1755 -  map 100% reduce 0%
2026-10-16 13:55:12 INFO  Job:1755 -  map 100% reduce 100%
2026-10-16 13:55:12 INFO  Job:1766 - Job job_1792157978563_0002 completed successfully
2026-10-16 13:55:12 INFO  Job:1773 - Counters: 54
        File System Counters
                FILE: Number of bytes read=56
                FILE: Number of bytes written=829443
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=238
                HDFS: Number of bytes written=26
                HDFS: Number of read operations=11
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=2
                Launched reduce tasks=1
                Rack-local map tasks=2
                Total time spent by all maps in occupied slots (ms)=4685
                Total time spent by all reduces in occupied slots (ms)=2363
                Total time spent by all map tasks (ms)=4685
                Total vcore-milliseconds taken by all map tasks=4685
                Total vcore-milliseconds taken by all reduce tasks=2363
                Total megabyte-milliseconds taken by all map tasks=4797440
                Total megabyte-milliseconds taken by all reduce tasks=2419712
        Map-Reduce Framework
                Map input records=2
                Map output records=4
                Map output bytes=42
                Map output materialized bytes=62
                Input split bytes=212
                Combine input records=4
                Combine output records=4
                Reduce input groups=3
                Reduce shuffle bytes=62
                Reduce input records=4
                Reduce output records=3
                Spilled Records=8
                Shuffled Maps =2
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=137
                CPU time spent (ms)=1370
                Physical memory (bytes) snapshot=763363328
                Virtual memory (bytes) snapshot=8064843776
                Total committed heap usage (bytes)=620232704
                Peak Map Physical memory (bytes)=298905600
                Peak Map Virtual memory (bytes)=2686464000
                Peak Reduce Physical memory (bytes)=210636800
                Peak Reduce Virtual memory (bytes)=2693156864
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=26
        File Output Format Counters
                Bytes Written=26
bash-4.2$
```

Demo: Container Local Word Count

```
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker compose up -d
[+] up 4/4
 ✓ Container datanode        Running
 ✓ Container resourcemanager Running
 ✓ Container nodemanager     Running
 ✓ Container namenode        Running
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker cp ../input.txt namenode:/tmp/
>> docker cp ../mapper.py namenode:/tmp/
>> docker cp ../reducer.py namenode:/tmp/
Successfully copied 86B (transferred 2.05kB) to namenode:/tmp/
Successfully copied 363B (transferred 2.05kB) to namenode:/tmp/
Successfully copied 682B (transferred 2.56kB) to namenode:/tmp/
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker exec -it namenode bash
bash-4.2$ cd /tmp
bash-4.2$ cat input.txt | python3 mapper.py
MapReduce,1
is,1
cool,,1
MapReduce,1
does,1
parallel,1
tasks,,1
Hadoop,1
is,1
fun.,1
Use,1
Hadoop,1
MapReduce.,1
bash-4.2$ cat input.txt | python3 mapper.py | sort -t',' -k1 | python3 reducer.py
Hadoop,2
MapReduce,2
MapReduce.,1
Use,1
does,1
fun.,1
is,2
parallel,1
bash-4.2$
```

Hadoop MapReduce Word Count

```
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker compose up -d
[+] up 4/4
 ✓ Container nodemanager     Running                                                    0.0s
 ✓ Container namenode        Running                                                    0.0s
 ✓ Container datanode        Running                                                    0.0s
 ✓ Container resourcemanager Running                                                    0.0s
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker cp ../big.txt namenode:/tmp/
>> docker cp ../mapper.py namenode:/tmp/
>> docker cp ../reducer.py namenode:/tmp/
Successfully copied 3.43kB (transferred 5.12kB) to namenode:/tmp/
Successfully copied 363B (transferred 2.05kB) to namenode:/tmp/
Successfully copied 682B (transferred 2.56kB) to namenode:/tmp/
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker exec -it namenode bash
bash-4.2$ cd /tmp
bash-4.2$ hdfs dfs -mkdir -p /user/root/input
bash-4.2$ hdfs dfs -put -f /tmp/big.txt /user/root/input/
bash-4.2$ hadoop jar /opt/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar \
>   -files /tmp/mapper.py,/tmp/reducer.py \
>   -input /user/root/input/big.txt \
>   -output /user/root/output_count \
>   -mapper "python3 mapper.py" \
>   -reducer "python3 reducer.py"
packageJobJar: [/tmp/hadoop-unjar4107888239906061721/] [] /tmp/streamjob2753877013116971095.jar tmpDir=null
2026-10-16 20:14:45 INFO  DefaultNoHARMFailoverProxyProvider:64 - Connecting to ResourceManager at resourcemanager/172.19.0.5:8032
2026-10-16 20:14:46 INFO  DefaultNoHARMFailoverProxyProvider:64 - Connecting to ResourceManager at resourcemanager/172.19.0.5:8032
2026-10-16 20:14:46 INFO  JobResourceUploader:907 - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1
792176943025_0011
2026-10-16 20:14:47 INFO  FileInputFormat:266 - Total input files to process : 1
2026-10-16 20:14:47 INFO  JobSubmitter:202 - number of splits:2
2026-10-16 20:14:47 INFO  JobSubmitter:298 - Submitting tokens for job: job_1792176943025_0011
2026-10-16 20:14:47 INFO  JobSubmitter:299 - Executing with tokens: []
2026-10-16 20:14:47 INFO  Configuration:2854 - resource-types.xml not found
2026-10-16 20:14:47 INFO  ResourceUtils:476 - Unable to find 'resource-types.xml'.
2026-10-16 20:14:47 INFO  YarnClientImpl:338 - Submitted application application_1792176943025_0011
2026-10-16 20:14:47 INFO  Job:1682 - The url to track the job: http://resourcemanager:8088/proxy/application_1792176943025_0011/
2026-10-16 20:14:47 INFO  Job:1727 - Running job: job_1792176943025_0011
2026-10-16 20:14:54 INFO  Job:1748 - Job job_1792176943025_0011 running in uber mode : false
2026-10-16 20:14:54 INFO  Job:1755 -  map 0% reduce 0%
2026-10-16 20:15:01 INFO  Job:1755 -  map 50% reduce 0%
2026-10-16 20:15:02 INFO  Job:1755 -  map 100% reduce 0%
2026-10-16 20:15:07 INFO  Job:1755 -  map 100% reduce 100%
```

```
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker exec -it namenode bash
bash-4.2$ cd /tmp
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=5148
        File Output Format Counters
                Bytes Written=3001
2026-10-16 20:15:07 INFO  StreamJob:1029 - Output directory: /user/root/output_count
bash-4.2$ hdfs dfs -cat /user/root/output_count/part-00000 | head -n 30
(for,1
1888--I,1
Adler.,1
All,1
And,1
As,1
Atkinson,1
Baker,2
Beyond,1
Bohemian,1
But,1
From,1
Grit,1
He,6
His,1
Holland.,1
Holmes,3
I,12
In,1
Irene,2
It,1
My,2
Odessa,1
One,1
Sherlock,1
Street.,1
Study,1
They,1
To,2
Trepoff,1
bash-4.2$
```

Distinct Word

```
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker compose up -d
[+] up 4/4
 ✓ Container resourcemanager Running                                                0.0s
 ✓ Container nodemanager     Running                                                0.0s
 ✓ Container namenode        Running                                                0.0s
 ✓ Container datanode        Running                                                0.0s
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker cp ../mapper_distinct.py namenode:/tmp/
>> docker cp ../reducer_distinct.py namenode:/tmp/
Successfully copied 381B (transferred 2.05kB) to namenode:/tmp/
Successfully copied 320B (transferred 2.05kB) to namenode:/tmp/
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker exec -it namenode bash
bash-4.2$ cd /tmp
bash-4.2$ hadoop jar /opt/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar \
>   -files /tmp/mapper_distinct.py,/tmp/reducer_distinct.py \
>   -input /user/root/input/big.txt \
>   -output /user/root/output_distinct \
>   -mapper "python3 mapper_distinct.py" \
>   -reducer "python3 reducer_distinct.py"
packageJobJar: [/tmp/hadoop-unjar3287910854562725404/] [] /tmp/streamjob6632054154986815385.jar tmpDir=null
2026-10-16 20:20:25 INFO  DefaultNoHARMFailoverProxyProvider:64 - Connecting to ResourceManager at resourcemanager/172.19.0.5:8032
2026-10-16 20:20:25 INFO  DefaultNoHARMFailoverProxyProvider:64 - Connecting to ResourceManager at resourcemanager/172.19.0.5:8032
2026-10-16 20:20:26 INFO  JobResourceUploader:907 - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_17921
76943025_0012
2026-10-16 20:20:26 INFO  FileInputFormat:266 - Total input files to process : 1
2026-10-16 20:20:26 INFO  JobSubmitter:202 - number of splits:2
2026-10-16 20:20:26 INFO  JobSubmitter:298 - Submitting tokens for job: job_1792176943025_0012
2026-10-16 20:20:26 INFO  JobSubmitter:299 - Executing with tokens: []
2026-10-16 20:20:27 INFO  Configuration:2854 - resource-types.xml not found
2026-10-16 20:20:27 INFO  ResourceUtils:476 - Unable to find 'resource-types.xml'.
2026-10-16 20:20:27 INFO  YarnClientImpl:338 - Submitted application application_1792176943025_0012
2026-10-16 20:20:27 INFO  Job:1682 - The url to track the job: http://resourcemanager:8088/proxy/application_1792176943025_0012/
2026-10-16 20:20:27 INFO  Job:1727 - Running job: job_1792176943025_0012
2026-10-16 20:20:35 INFO  Job:1748 - Job job_1792176943025_0012 running in uber mode : false
2026-10-16 20:20:35 INFO  Job:1755 -  map 0% reduce 0%
2026-10-16 20:20:41 INFO  Job:1755 -  map 100% reduce 0%
2026-10-16 20:20:45 INFO  Job:1755 -  map 100% reduce 100%
2026-10-16 20:20:45 INFO  Job:1766 - Job job_1792176943025_0012 completed successfully
2026-10-16 20:20:45 INFO  Job:1773 - Counters: 54
        File System Counters
                FILE: Number of bytes read=5807
                FILE: Number of bytes written=851781
```

```
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker exec -it namenode bash
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=5148
        File Output Format Counters
                Bytes Written=2500
2026-10-16 20:20:45 INFO  StreamJob:1029 - Output directory: /user/root/output_distinct
bash-4.2$ hdfs dfs -cat /user/root/output_distinct/part-00000 | head -n 30
1888i
a
abandoned
abhorrent
absorb
accomplished
account
actions
activity
adjusted
adler
admirable
admirably
admit
again
against
akin
all
alternating
always
ambition
among
and
any
around
as
associated
at
atkinson
attention
bash-4.2$
```

Combiner

```
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker compose up -d
[+] up 4/4
 ✓ Container namenode        Running                                                                    0.0s
 ✓ Container datanode        Running                                                                    0.0s
 ✓ Container resourcemanager Running                                                                    0.0s
 ✓ Container nodemanager     Running                                                                    0.0s
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker cp ../mapper_wc.py namenode:/tmp/
>> docker cp ../combiner_wc.py namenode:/tmp/
>> docker cp ../reducer_wc.py namenode:/tmp/
Successfully copied 211B (transferred 2.05kB) to namenode:/tmp/
Successfully copied 549B (transferred 2.56kB) to namenode:/tmp/
Successfully copied 553B (transferred 2.56kB) to namenode:/tmp/
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker exec -it namenode bash
bash-4.2$ cd /tmp
bash-4.2$ hadoop jar /opt/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar \
>   -files /tmp/mapper_wc.py,/tmp/combiner_wc.py,/tmp/reducer_wc.py \
>   -input /user/root/input/big.txt \
>   -output /user/root/output_wc_combiner \
>   -mapper "python3 mapper_wc.py" \
>   -combiner "python3 combiner_wc.py" \
>   -reducer "python3 reducer_wc.py"
packageJobJar: [/tmp/hadoop-unjar6767526234887336808/] [] /tmp/streamjob5842518666938958451.jar tmpDir=null
2026-10-16 20:27:38 INFO  DefaultNoHARMFailoverProxyProvider:64 - Connecting to ResourceManager at resourcemanager/172.19.0.5:8032
2026-10-16 20:27:39 INFO  DefaultNoHARMFailoverProxyProvider:64 - Connecting to ResourceManager at resourcemanager/172.19.0.5:8032
2026-10-16 20:27:39 INFO  JobResourceUploader:907 - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_17921
76943025_0013
2026-10-16 20:27:39 INFO  FileInputFormat:266 - Total input files to process : 1
2026-10-16 20:27:40 INFO  JobSubmitter:202 - number of splits:2
2026-10-16 20:27:40 INFO  JobSubmitter:298 - Submitting tokens for job: job_1792176943025_0013
2026-10-16 20:27:40 INFO  JobSubmitter:299 - Executing with tokens: []
2026-10-16 20:27:40 INFO  Configuration:2854 - resource-types.xml not found
2026-10-16 20:27:40 INFO  ResourceUtils:476 - Unable to find 'resource-types.xml'.
2026-10-16 20:27:40 INFO  YarnClientImpl:338 - Submitted application application_1792176943025_0013
2026-10-16 20:27:41 INFO  Job:1682 - The url to track the job: http://resourcemanager:8088/proxy/application_1792176943025_0013/
2026-10-16 20:27:41 INFO  Job:1727 - Running job: job_1792176943025_0013
2026-10-16 20:27:48 INFO  Job:1748 - Job job_1792176943025_0013 running in uber mode : false
2026-10-16 20:27:48 INFO  Job:1755 -  map 0% reduce 0%
2026-10-16 20:27:54 INFO  Job:1755 -  map 50% reduce 0%
2026-10-16 20:27:55 INFO  Job:1755 -  map 100% reduce 0%
2026-10-16 20:28:00 INFO  Job:1755 -  map 100% reduce 100%
2026-10-16 20:28:00 INFO  Job:1766 - Job job_1792176943025_0013 completed successfully
```

```
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker exec -it namenode bash
bash-4.2$ cd /tmp
                Total committed heap usage (bytes)=731381760
                Peak Map Physical memory (bytes)=303706112
                Peak Map Virtual memory (bytes)=2687553536
                Peak Reduce Physical memory (bytes)=242585600
                Peak Reduce Virtual memory (bytes)=2692816896
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=5148
        File Output Format Counters
                Bytes Written=3119
2026-10-16 20:28:00 INFO  StreamJob:1029 - Output directory: /user/root/output_wc_combiner
bash-4.2$ hdfs dfs -cat /user/root/output_wc_combiner/part-00000 | head -n 20
(for    1
1888--I 1
Adler,  1
Adler.  1
All     1
And     1
As      1
Atkinson        1
Baker   2
Beyond  1
Bohemian        1
But     1
From    1
Grit    1
He      6
His     1
Holland.        1
Holmes  3
Holmes, 1
I       12
bash-4.2$
```

K Means

```
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker compose up -d
[+] up 4/4
 ✓ Container nodemanager     Running                                                                    0.0s
 ✓ Container namenode        Running                                                                    0.0s
 ✓ Container datanode        Running                                                                    0.0s
 ✓ Container resourcemanager Running                                                                    0.0s
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker cp ../iris.data namenode:/tmp/
>> docker cp ../mapper_kmeans.py namenode:/tmp/
>> docker cp ../reducer_kmeans.py namenode:/tmp/
Successfully copied 4.55kB (transferred 6.14kB) to namenode:/tmp/
Successfully copied 979B (transferred 2.56kB) to namenode:/tmp/
Successfully copied 981B (transferred 2.56kB) to namenode:/tmp/
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker exec -it namenode bash
bash-4.2$ cd /tmp
bash-4.2$ hdfs dfs -put -f /tmp/iris.data /user/root/input/
bash-4.2$ hadoop jar /opt/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar \
>   -files /tmp/mapper_kmeans.py,/tmp/reducer_kmeans.py \
>   -input /user/root/input/iris.data \
>   -output /user/root/output_kmeans \
>   -mapper "python3 mapper_kmeans.py" \
>   -reducer "python3 reducer_kmeans.py"
packageJobJar: [/tmp/hadoop-unjar3977698968482477684/] [] /tmp/streamjob7057874782311674400.jar tmpDir=null
2026-10-16 20:33:03 INFO  DefaultNoHARMFailoverProxyProvider:64 - Connecting to ResourceManager at resourcemanager/172.19.0.5:8032
2026-10-16 20:33:04 INFO  DefaultNoHARMFailoverProxyProvider:64 - Connecting to ResourceManager at resourcemanager/172.19.0.5:8032
2026-10-16 20:33:04 INFO  JobResourceUploader:907 - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_17921
76943025_0014
2026-10-16 20:33:04 INFO  FileInputFormat:266 - Total input files to process : 1
2026-10-16 20:33:05 INFO  JobSubmitter:202 - number of splits:2
2026-10-16 20:33:05 INFO  JobSubmitter:298 - Submitting tokens for job: job_1792176943025_0014
2026-10-16 20:33:05 INFO  JobSubmitter:299 - Executing with tokens: []
2026-10-16 20:33:05 INFO  Configuration:2854 - resource-types.xml not found
2026-10-16 20:33:05 INFO  ResourceUtils:476 - Unable to find 'resource-types.xml'.
2026-10-16 20:33:05 INFO  YarnClientImpl:338 - Submitted application application_1792176943025_0014
2026-10-16 20:33:05 INFO  Job:1682 - The url to track the job: http://resourcemanager:8088/proxy/application_1792176943025_0014/
2026-10-16 20:33:05 INFO  Job:1727 - Running job: job_1792176943025_0014
2026-10-16 20:33:12 INFO  Job:1748 - Job job_1792176943025_0014 running in uber mode : false
2026-10-16 20:33:12 INFO  Job:1755 -  map 0% reduce 0%
2026-10-16 20:33:18 INFO  Job:1755 -  map 50% reduce 0%
2026-10-16 20:33:19 INFO  Job:1755 -  map 100% reduce 0%
2026-10-16 20:33:24 INFO  Job:1755 -  map 100% reduce 100%
2026-10-16 20:33:24 INFO  Job:1766 - Job job_1792176943025_0014 completed successfully
```

```
PS D:\Coding\Learning\CloudComputing\Assignment08\hadoop_mapReduce> docker exec -it namenode bash
bash-4.2$ cd /tmp
                Map output bytes=1650
                Map output materialized bytes=1962
                Input split bytes=186
                Combine input records=0
                Combine output records=0
                Reduce input groups=3
                Reduce shuffle bytes=1962
                Reduce input records=150
                Reduce output records=3
                Spilled Records=300
                Shuffled Maps =2
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=146
                CPU time spent (ms)=1940
                Physical memory (bytes) snapshot=801239040
                Virtual memory (bytes) snapshot=8070291456
                Total committed heap usage (bytes)=643825664
                Peak Map Physical memory (bytes)=301322240
                Peak Map Virtual memory (bytes)=2687602688
                Peak Reduce Physical memory (bytes)=245063680
                Peak Reduce Virtual memory (bytes)=2695405568
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=6827
        File Output Format Counters
                Bytes Written=201
2026-10-16 20:33:24 INFO  StreamJob:1029 - Output directory: /user/root/output_kmeans
bash-4.2$ hdfs dfs -cat /user/root/output_kmeans/part-00000
C1      Updated Candidate Point: (5.08, 3.55)   Total Points Assigned: 39
C2      Updated Candidate Point: (5.59, 2.83)   Total Points Assigned: 62
C3      Updated Candidate Point: (6.78, 2.94)   Total Points Assigned: 49
bash-4.2$
```